EE/CE 354/361: Introduction to Probability and Statistics

Homework Assignment 1: Probability Basics

Instructions

- **Attempt all questions.** Only **7 randomly chosen** questions will be graded.
- Show all working clearly.
- State assumptions where necessary.
- Total Points: **70**

Q1 (10 points)

Let the sample space be $S = \{1, 2, 3, 4, 5, 6\}$, corresponding to the outcome of rolling a fair six-sided die. Let events A and B be:

$$A = \{2, 4, 6\}, \quad B = \{1, 2, 3, 4\}$$

Describe the following sets by listing their elements:

1. $A \cup B$
2. $A \cap B$
3. A^c

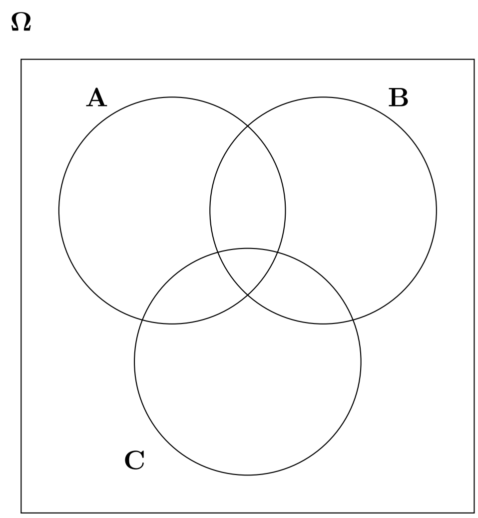
Q2 (10 points)

A deck of 52 playing cards is dealt. Let

$$A = \{\text{the card is a heart}\}, \quad B = \{\text{the card is a face card (J, Q, K)}\}, \quad C = \{\text{the card is an ace}\}$$

Express each of the following regions in set notation (using unions, intersections, and complements; **do not calculate probabilities**). Then, shade the corresponding regions on a Venn diagram. A sample diagram is provided below.

1. The event that the card is either a heart or a face card but not both.
2. The event that the card is neither a heart nor an ace.
3. The event that the card is either an ace or a face card, but not both, and also not a heart.



Q3 (10 points)

A box contains 3 red balls, 2 blue balls, and 1 green ball. One ball is chosen at random.

1. Describe the sample space.
2. Assign probabilities to each outcome.
3. What is the probability of choosing a non-red ball?

Q4 (10 points)

A factory produces 60% of its products in Plant A and 40% in Plant B.

- 5% of Plant A's products are defective.
- 10% of Plant B's products are defective.

If a product is chosen at random and found to be defective, what is the probability that it came from Plant B?

Q5 (10 points)

A medical test is designed to detect a disease that occurs in 1% of the population.

- If a person has the disease, the test is positive with probability 0.95.
- If a person does not have the disease, the test is (falsely) positive with probability 0.02.

A person is chosen at random and tests positive. What is the probability that the person actually has the disease?

Q6 (10 points)

Consider an experiment consisting of two successive rolls of a fair 4-sided die. Consider the following events in this experiment:

A = {The result of first roll is less than 3}

B = {The result of second roll is greater than 2}

C = {The sum of two rolls is greater than 6}

Justify (by calculating appropriate probabilities) whether the following statements are true or false:

(a) A and C are independent events.

(b) A, B, and C are independent events.

(c) A and B are not conditionally independent with respect to event C.

Q7 (10 points)

Consider two consecutive rolls of a fair 6-sided die. Are the following events independent?

A = {1st roll results in 1, 2, or 4}

B = {The result of both the rolls is the same}

Justify your answer.

Q8 (10 points)

How many different 4-digit numbers can be formed using the digits $\{1, 2, 3, 4, 5\}$ if digits can be repeated?

Q9 (10 points)

A password consists of 6 characters chosen from the 26 uppercase English letters (A–Z), with repetition allowed.

1. How many possible passwords are there in total?
2. How many possible passwords contain at least one vowel (A, E, I, O, U)?
3. What is the probability that a randomly chosen password contains at least one vowel?

Q10 (10 points)

A university committee of 5 is to be formed from 8 professors and 6 students.

1. In how many ways can the committee be formed if it must contain at least 2 students?
2. What is the probability that a randomly chosen committee contains exactly 3 professors and 2 students?

Q11 (10 points)

Fifteen dots are evenly placed on the perimeter of a circle. How many three dots can we pick from these 15 that do **not** form an equilateral triangle? Note that the order of dots selected does not matter. In other words, selecting dots {1, 2, 3} is the same as selecting dots {3, 1, 2}.

Q12 (10 points)

In a city with one hundred taxis, 1 is blue and 99 are green. A witness observes a hit-and-run by a taxi at night and recalls that the taxi was blue, so the police arrest the blue taxi driver who was on duty that night. The driver proclaims his innocence and hires you to defend him in court. You hire a scientist to test the witness's ability to distinguish blue and green taxis under conditions similar to the night of accident. The data suggests that the witness sees blue cars as blue 99% of the time and green cars as blue 2% of the time. What is the probability that taxi was actually blue given the witness's statement?